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The Family Quality of Life of Children With Specific Learning Disabilities, Attention-Deficit/Hyperactivity Disorder, and Their Co-Occurrence: A Cross-Sectional Study

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Abstract

Research on Family Quality of Life (FQoL) has increasingly focused on the impact of learning disabilities on families. However, there is a lack of comprehensive understanding of how Specific Learning Disabilities (SpLD) or Attention-Deficit Hyperactivity/Disorder (ADHD) affect executive function (EF) deficits in children and FQoL across different household income levels. The current study compared the FQoL ratings from caregivers of three groups of Hong Kong Chinese children with learning disabilities (SpLD $n = 107$; ADHD $n = 43$; SpLD & ADHD $n = 67$, no typically developing control group was included). Results revealed that group classification of learning disabilities was related to two domains of FQoL (family interaction and parenting) through the child's EF. Children with co-occurring SpLD and ADHD exhibit relatively stable EF deficits across income levels, maintaining higher EF deficits than those with SpLD alone and showing less variation than those with ADHD, who demonstrate a significant decrease in EF deficits at higher household income. Executive function deficits were associated with lower family interaction and parenting ratings across group and income levels. The findings enhance our understanding of FQoL among children with SpLD, ADHD, and their co-occurrence, highlighting the need for additional support for these families.

Keywords

ADHD, caretaker, comorbidity, executive function, family quality of life, socioeconomic status, SpLD

Family Quality of Life (FQoL) represents a dynamic sense of collective well-being shaped by individual family members, reflecting the interplay between individual and family-level needs (Wang & Kober, 2011). Rooted in the family systems theory, FQoL recognizes families as intricate social entities where members mutually influence each other's behaviors (Kerr, 1981). This interconnectedness means that one family member's experiences ripple through the entire family system, impacting the well-being of others (Samuel et al., 2011). In the disability field, the presence of special educational needs in children and their families' reactions takes on particular significance as they can profoundly affect their FQoL (Wang et al., 2004; Wang & Kober, 2011). The current study aims to shed light on the complex relationship between special educational needs diagnoses and FQoL, with a specific focus on the roles of the child's executive functions (EF) and family socioeconomic status (SES), especially among children with Specific Learning Disabilities (SpLD), Attention-Deficit/Hyperactivity Disorder (ADHD), and their co-occurrence.

The Impact of SpLD, ADHD, and Their Co-Occurrence on FQoL

Children are diagnosed with ADHD through a combination of behavioral assessments, clinical interviews, and standardized rating scales that evaluate symptoms such as inattention, hyperactivity, and impulsivity. According to *Diagnostic and Statistical Manual of Mental Disorders* (5th ed.; *DSM-5*; American Psychiatric Association, 2013) criteria, these symptoms must be present in multiple settings and must significantly impair functioning (Harris, 2024). ADHD is one of the most common psychiatric disorders in

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childhood, affecting approximately 6.4% of children and adolescents in Hong Kong (Liu et al., 2018). On the contrary, Specific Learning Disabilities (SpLD) are characterized by unexpected but significant difficulties in specific learning areas, such as reading and mathematics, and are diagnosed using standardized assessments and clinical evaluations. SpLD affects up to 12.6% of elementary and secondary students in Hong Kong (D. Chan et al., 2007). Caregivers of children with learning disabilities like ADHD or SpLD often face psychological, social, and managerial challenges, which can have profound implications for FQoL (e.g., Dizdarevic et al., 2022; Xiang et al., 2009).

Previous research consistently reveals a negative correlation between FQoL and the severity of behavioral problems in children with ADHD, a pattern observed both in Western contexts and in Hong Kong (e.g., Armstrong et al., 2014; Cussen et al., 2012; Romaniuk et al., 2022; Xiang et al., 2009). Parents of children with ADHD, particularly those of children with the inattention and combined subtypes, often experience depression and anxiety due to the child's ADHD symptoms and their own ineffective coping strategies, which can lower FQoL and family functioning (Cussen et al., 2012; Kashdan et al., 2004; Smit et al., 2021). Despite the high prevalence of SpLD among school-age children, only recently has a single study examined its impact on FQoL (Xu et al., 2025). The findings from this study revealed that the challenges associated with SpLD, including academic struggles and emotional and behavioral difficulties, had a significant impact on parental stress and a negative effect on the child's self-esteem and overall FQoL (Xu et al., 2025). However, given that this is a single study, the existing body of research on the FQoL in children with learning disabilities remains limited, particularly in Chinese contexts.

In addition, there is a distinct lack of research on the FQoL of children with comorbid ADHD and SpLD. Comorbidity refers to the presence of multiple special educational needs diagnoses. In Hong Kong, around 25% of families recruited from primary schools reported comorbid developmental disabilities in their children (R. C. H. Chan et al., 2020). Studies have estimated that up to 30% of individuals with ADHD also present with dyslexia (Hendren et al., 2018; Nelson & Gregg, 2012). Children with comorbid ADHD and SpLD may exhibit more pronounced difficulties across various domains of functioning, placing greater demands on caregivers (Hendren et al., 2018; Karande & Venkataraman, 2013; Taurines et al., 2010; Willcutt et al., 2001). Severe psychosocial difficulties in the child are likely to elicit strict parenting in response, resulting in tension in family relationships (R. C. H. Chan et al., 2020). Given the high comorbidity rate between ADHD and SpLD, studying these conditions together, rather than in isolation, is imperative.

The Role of Executive Function and SES

Families caring for children with SpLD or ADHD frequently confront challenges linked to deficits in the child's EF (Frazier et al., 2022; Rapport et al., 2013). EF is a sophisticated cognitive process that coordinates thoughts, emotions, and actions, and it includes components such as working memory, inhibitory control, and cognitive flexibility (Elliott, 2003). These EF components are crucial for goal-directed behaviors and highly correlated with daily functions and academic success (Dajani et al., 2016). Deficits in EFs in children with ADHD or SpLD significantly impact their social, emotional, and academic functioning. Research indicates that children with ADHD exhibit impairments in EFs which correlate with difficulties in social interactions and emotional regulation, leading to increased behavioral issues and peer rejection (Chiang & Gau, 2014). Similarly, children with SpLD often face challenges in EFs that manifest in academic tasks such as reading and writing, which can exacerbate feelings of frustration and low self-esteem (Zuppardo et al., 2023). Moreover, severe executive dysfunction is linked with a range of challenging behaviors, including property destruction, self-injury, or aggression (Allen et al., 2017; Frazier et al., 2022; Takacs & Kassai, 2019), potentially increasing the family's financial burden due to costs associated with damages or therapeutic interventions. These dysfunctions can significantly impact overall family dynamics and the caregivers' personal well-being. In summary, deficits in EFs can affect various aspects of FQoL, such as family interaction and emotional well-being, which can tax the family's coping mechanisms by inducing additional stress and potential disruptions to their daily life (Frazier et al., 2022).

While comorbid SpLD and ADHD may result in unique combinations of executive dysfunctions, they do not necessarily indicate more profound deficits compared with single diagnoses due to their shared symptoms (Fernández-Andrés et al., 2019; Moura et al., 2016). To clarify this point, recent studies have shown that comorbid SpLD and ADHD may result in unique combinations of executive dysfunctions (Bental & Tirosh, 2007; Poon & Ho, 2014; Willcutt et al., 2005), which may suggest a compounded effect on FQoL when multiple learning disabilities are present. However, this connection still requires empirical research to substantiate. The close relationships between learning disabilities and EF, and between EF and behavioral difficulties in children, hint at potential connections between these variables. The nature of these connections, whether causal or correlational, remains largely unexplored and warrants further investigation.

Furthermore, the influence of family SES on these dynamics between EF and FQoL is critical but not yet comprehensively understood. Socioeconomic status encompasses

factors such as parents' education levels, occupations, and household income. It can be related to physical well-being, one aspect of FQoL measuring the subjective availability of resources to support the family's daily needs. High-SES families may have a higher rate of marital harmony and employ less coercive discipline than low-SES families (Hartley et al., 2012; Wang et al., 2004). In particular, household income plays a critical role in child development, as poverty has been associated with long-term stress and deficits in EF due to factors such as lack of nutrition and unsafe home environment (Blair, 2010; Duncan et al., 2017). These suggest moderating effects of financial income on children's EF and family well-being. Low-income families of children with ADHD or SpLD may have fewer available resources to support their children's education, intensifying their vulnerabilities (Guajardo et al., 2009). Conversely, investing financial resources into supporting the children's diverse educational needs may diminish the family's overall material resources. For instance, parents of children with learning disabilities may choose to forego full-time work to spend more time with their children, thus reducing the household income. Although high-income families tend to possess more resources and access to information to address challenges associated with learning disabilities, how this interacts with executive dysfunctions observed in children with ADHD or SpLD remains inadequately explored. Understanding these intricate relationships is pivotal for informed interventions and policy support.

Study Aims

Given these research gaps, our study aimed to understand the FQoL among families in Hong Kong with children diagnosed with ADHD, SpLD, or both. Specifically, the study sought to compare the FQoL of these families with that of the general population and explored how these conditions, in conjunction with SES, affect a child's EF and overall FQoL.

We anticipated that children with any of these conditions would have a lower FQoL than the general population, and those with both SpLD and ADHD would have an even lower FQoL due to their greater difficulties with EF (e.g., Karande & Venkataraman, 2013; Taurines et al., 2010). Thus, we hypothesized that EF would mediate the relationship between a child's SpLD/ADHD status and FQoL (Frazier et al., 2022; Poon & Ho, 2014), potentially revealing the mechanisms through which learning disabilities affect family well-being. In addition, we examined the moderating role of household income in the relationships between a child's SpLD/ADHD status, EF, and FQoL, expecting that it might influence various aspects of this mediation and offer critical insights for tailored interventions. We also hypothesized that children with both SpLD and ADHD from lower-income families might demonstrate disproportionately lower EF (Blair, 2010; Duncan et al., 2017), resulting in cumulative reductions in FQoL.

Method

Design and Participants

In this cross-sectional study, participants were adult caregivers of a child with SpLD, ADHD, or both SpLD and ADHD. Participants were recruited from ordinary primary schools in Hong Kong. The authors obtained the children's diagnostic information from the schoolteachers after the parents or caregivers consented. Children with ADHD or SpLD diagnoses were identified based on comprehensive psychological reports conducted by clinical or educational psychologists or psychiatrists and included multiple sources of information. These reports included results from standardized psychological assessments on students' abilities in reading and writing, such as *The Hong Kong Test of Specific Learning Difficulties in Reading and Writing for Primary Students—Third Edition* (HKT-P(III); Ho et al., 2016). ADHD-specific symptoms were assessed using tools like the Conners Continuous Performance Test (Conners et al., 2018), alongside other clinical evaluations and behavioral checklists, such as the Conners Parent and Teacher Rating Scales. Full-scale IQ was also evaluated. Participants were then grouped based on their child's special educational needs conditions (ADHD, SpLD, or Co-occurrence), regardless of other comorbidities.

All participants gave written informed consent before responding to the survey. Of the initial 325 responses, 217 were valid for analysis. Excluded responses were due to incomplete data or withdrawal of consent from participants. Data were analyzed from caregivers of three groups of Hong Kong Chinese children with learning disabilities based on certified special educational needs diagnoses (SpLD $n = 107$; ADHD $n = 43$; SpLD & ADHD $n = 67$; no typically developing control group was included). We conducted a power analysis of a three-group one-way analysis of variance (ANOVA) using GPower with input parameters of effect size (η^2) of 0.25, an alpha level of .05, and a power of .9, which yielded a sample size of 210. Therefore, we inferred that our data had sufficient power to detect the intended effects.

Measures

Demographic Questions. The questionnaire included demographic information including age, sex, relationship with the child, and parental education level. The child's special educational needs types were indicated on a list, including SpLD, ADHD, autism spectrum disorder, intellectual disabilities, developmental language disorder, social communication disorder, sensory impairments,

emotional disorders, or other conditions. Monthly household income was scored on a scale of 1 to 12 from "$10,000 or below" to "$50,001 or above" in Hong Kong dollars and was used as a proxy measure of SES.

Family Quality of Life. Participants reported their FQoL using The Beach Center Family Quality of Life Scale (BC-FQoL; Hoffman et al., 2006). This scale contains 25 items in five subscales: Family Interaction (e.g., *My family solves problems together*), Parenting (e.g., *Family members teach the child how to get along with others*), Emotional Well-being (e.g., *My family has the support we need to relieve stress*), Physical/Material Well-being (e.g., *My family gets dental care when needed*), and Disability-Related Support (e.g., *My family member with special needs has support to make friends*). Each item was rated on a Likert-type scale from 1 to 5 (*very dissatisfied* to *very satisfied*), with higher scores indicating greater FQoL. Based on previous literature (Xiang et al., 2009) and available local statistics, we included only the first four subscales (21 items) to understand the caregivers' satisfaction with their FQoL. The internal consistencies for the subscales were acceptable, with Cronbach's alpha ranging from .724 to .891 and McDonald's omega from .769 to .894.

Behavior Rating Inventory of Executive Function. An abbreviated 24-item version of the Behavior Rating Inventory of Executive Function (BRIEF) for children ages 5 to 18 was used to assess EF of children in the context of their home environments (LeJeune et al., 2010). The response categories were 1 = never, 2 = sometimes, and 3 = often, with total scores ranging from 24 to 72. The raw scores were converted to T-scores, with scores of 60 or above indicating elevated executive dysfunction. The Cronbach's alpha and McDonald's omega for this scale were both .905.

Procedure

Participants completed the questionnaire either on paper or electronically on the Qualtrics platform. Data were collected between May 2021 and February 2022. This study was reviewed and approved by the research ethics committee of the first author's institution.

Data Analysis

Household income and BRIEF T-scores were standardized into *z*-scores in the statistical models. We used one-way ANOVAs to test the hypothesis that parenting children with co-occurring SpLD and ADHD is associated with lower FQoL levels than parenting children with SpLD or ADHD alone. Significant group effects were followed up using Tukey's post hoc multiple comparison tests to identify pairwise differences. To examine the mediating effect of EF and the moderating effect of household income, we ran a conditional process model using Model 7 of the SPSS Macro Program PROCESS (version 2.13) in SPSS (Hayes, 2013). The predictor variable was group classification (SpLD, ADHD, or Co-occurrence), and dummy variable coding was used as there were three groups. The Co-occurrence group was set as the reference level, such that the mediation and moderation effects showed comparisons of the single diagnosis groups (SpLD or ADHD) relative to their co-occurrence. The child's EF was tested as a mediator between group classification and FQoL using bias-corrected 95% confidence interval (parametric bootstrapping of 5,000 samples), with household income as the moderating variable on the mediations (see Figure 1A). In the moderated mediation models, unstandardized β of the conditional indirect effects were presented. An absence of zero within the confidence intervals or p values of .05 or below were taken as significant effects for hypothesis testing.

Results

Descriptive Statistics

Demographic information for the three groups is shown in Table 1. There were no significant differences in demographic variables between the three groups, except for the number of other co-occurring special educational needs. Post hoc comparisons showed that the ADHD group reported more comorbid special educational needs than the SpLD group (0.7 and 0.3, respectively, $p = .014$). Table 2 shows the bivariate correlations of the continuous variables in the study. In line with previous reports, subscales of FQoL were negatively related to the child's executive dysfunctions (rs = −.322 to −.145), while all subscales of FQoL except for Parenting are positively related to household income (rs = .137 to .167).

Group Differences

The mean BC-FQoL scores were below 3 out of 5 and the mean BRIEF T-scores were above 60 for all three groups (see Figure 2), suggesting that the participants were generally dissatisfied with their FQoL and reported elevated executive dysfunction in their children with learning disabilities. Higher FQoL scores represent more positive outcomes, which is opposite to the directionality of EF scores, where higher scores indicate more dysfunction. Results of the one-way ANOVAs showed small to medium group differences in EF scores and two FQoL domains, namely, family interaction and parenting. Post hoc analyses showed that the SpLD group displayed better EF, that is, lower T-scores

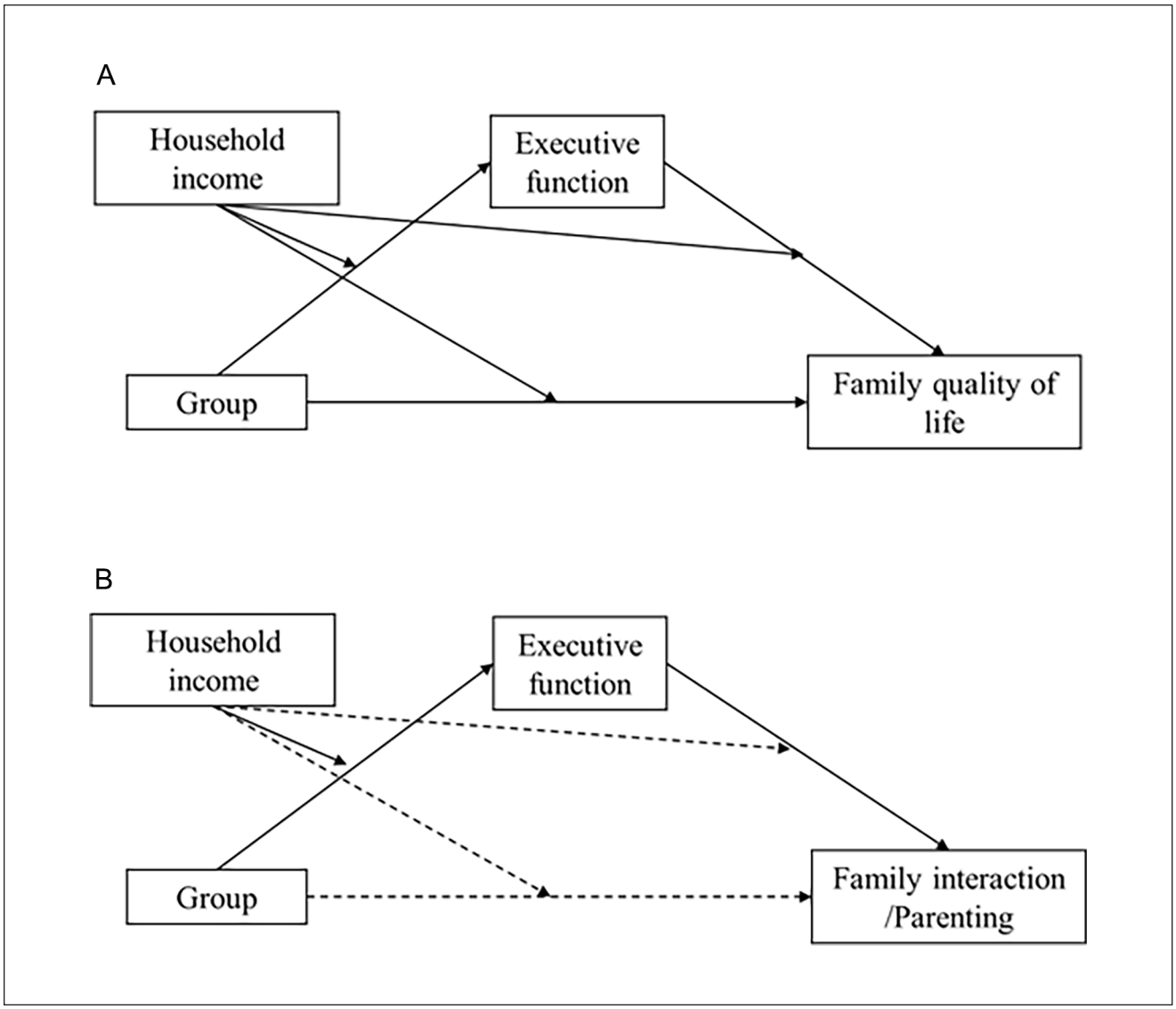


Figure 1. The (A) Conceptual Diagram and (B) Final Results of the Models, With Dotted Lines Indicating Nonsignificant Paths.

in the BRIEF scale ($M = 62.6$), than both the Co-occurrence group ($M = 68.8$; Cohen's $d = 0.70$, $p = .021$) and the ADHD group ($M = 67.3$; Cohen's $d = 0.48$, $p < .001$), with medium to large effect sizes. No significant difference in EF was observed between the Co-occurrence and the ADHD groups ($p = .674$). Regarding the family interaction domain of FQoL, there was a trend for the SpLD group to score higher than the Co-occurrence group (Cohen's $d = 0.41$, $p = .058$). Similarly, the SpLD group had marginally higher parenting scores than the Co-occurrence group (Cohen's $d = 0.38$, $p = .073$) and the ADHD group (Cohen's $d = 0.50$, $p = .085$). There was no difference between the Co-occurrence and ADHD groups in the family interaction ($p = .954$) and the parenting ($p = .978$) domains.

Mediation and Moderation Effects

As the ANOVA results showed potential group differences for the family interaction and parenting BC-FQoL subscales, we conducted the moderated mediation analyses for these two outcomes separately to investigate the underlying mechanisms. The hypothesized model tested whether household income moderated the purported group differences in EF, affecting FQoL. The following paragraphs first outline the effects on BRIEF T-scores before examining the overall model effects on FQoL.

The model results indicated lower BRIEF T-scores for the SpLD group compared with the Co-occurrence group ($\beta = -0.65$, $p < .001$), but no main effect difference between the ADHD and Co-occurrence groups ($\beta = -0.21$, $p = .26$).

Table 1. Demographic Characteristics of the Participants Split by SpLD/ADHD Status.

Characteristic	SpLD ($n = 107$)	ADHD ($n = 43$)	SpLD and ADHD ($n = 67$)	Test value	p value
Child age: years, *M* (*SD*)	10.8 (1.37)	10.7 (1.46)	10.6 (1.24)	0.94	.624
Child sex				4.84	.089
Male	66 (61.7%)	33 (76.7%)	50 (74.6%)		
Female	41 (38.3%)	10 (23.3%)	17 (25.4%)		
Number of other special educational needs	0.30 (0.70)	0.70 (1.06)	0.54 (0.91)	9.24	.010
Relationship with child				3.30	.509
Father	19 (17.8%)	7 (16.3%)	7 (10.4%)		
Mother	80 (74.8%)	35 (81.4%)	56 (83.6%)		
Guardian	8 (7.48%)	1 (2.32%)	4 (5.97%)		
Marital status of parents				1.53	.466
Not yet married	10 (9.3%)	0	5 (7.46%)		
Widowed	2 (1.87%)	1 (2.32%)	1 (1.49%)		
Separated	4 (3.74%)	2 (4.65%)	0		
Divorced	6 (5.61%)	4 (9.30%)	4 (5.97%)		
Married	81 (75.7%)	36 (83.7%)	57 (85.1%)		
Paternal education level				9.59	.143
Primary school or below	7 (6.9%)	5 (12.5%)	5 (7.7%)		
Secondary school	86 (85.2%)	30 (75.0%)	50 (76.9%)		
College and University	5 (5.0%)	5 (12.5%)	10 (15.4%)		
Master's degree or above	3 (3.0%)	0	0		
Maternal education level				6.14	.407
Primary school or below	10 (9.7%)	8 (19.5%)	7 (10.6%)		
Secondary school	80 (77.7%)	26 (63.4%)	47 (71.2%)		
College and University	12 (11.7%)	6 (14.6%)	12 (18.1%)		
Master's degree or above	1 (1.0%)	1 (2.4%)	0		
Household Income (Hong Kong $)				0.56	.758
< $5,000 or below	5 (4.67%)	1 (2.32%)	0		
$5,001–$10,000	8 (7.48%)	6 (13.9%)	6 (8.96%)		
$10,001–$15,000	12 (11.2%)	8 (18.6%)	12 (17.9%)		
$15,001–$20,000	27 (25.2%)	7 (16.3%)	15 (22.4%)		
$20,001–$25,000	22 (20.6%)	7 (16.3%)	11 (16.4%)		
$25,001–$30,000	8 (7.48%)	5 (11.6%)	10 (14.9%)		
$30,001–$35,000	6 (5.61%)	4 (9.30%)	4 (5.97%)		
$35,001–$40,000	7 (6.54%)	1 (2.32%)	2 (2.99%)		
$40,001–$45,000	3 (2.80%)	2 (4.65%)	2 (2.99%)		
$45,001–$50,000	1 (0.93%)	1 (2.32%)	1 (1.49%)		
$50,001–$55,000	3 (2.80%)	1 (2.32%)	2 (2.99%)		
$55,001 or above	5 (4.67%)	0	2 (2.99%)		

Note. For continuous measures, means (standard deviations) are shown. For count measures, the numbers of occurrences (% within group) are shown. The analysis of paternal and maternal education levels was conducted after excluding participants for whom no information was available.
SpLD = specific learning disabilities; ADHD = attention-deficit/hyperactivity disorder.

Household income did not predict BRIEF T-scores ($\beta = 0.02$, $p = .89$). However, household income moderated the difference between ADHD and Co-occurrence groups in their BRIEF T-scores ($\beta = -0.50$, $p < .05$). The same interaction between group and household income was not seen in the comparison between SpLD and Co-occurrence groups ($\beta = -0.03$, $p = .85$). Figure 3 shows how the group differences in BRIEF T-scores depended on household income. When household income was at or below the mean, both the ADHD and Co-occurrence groups had elevated executive dysfunction compared with the SpLD group. When household income was high, the ADHD group had lower BRIEF T-scores such that the gap between the ADHD and SpLD groups narrowed, but the BRIEF T-scores of the Co-occurrence group remained high.

The child's BRIEF T-scores had a small but reliable negative effect on Family Interaction ($\beta = -0.18$, $p < .001$) and Parenting ($\beta = -0.16$, $p < .001$), indicating that more

Table 2. Bivariate Correlations of the Variables in the Model.

Variables	1	2	3	4	5	6	7
1. Child age	—						
2. Family interaction	−.017	—					
3. Parenting	.049	.745***	—				
4. Emotional well-being	.069	.624***	.732***	—			
5. Physical well-being	.062	.507***	.625***	.582***	—		
6. Overall family quality of life	.044	.856***	.920***	.835***	.794***	—	
7. Executive dysfunction	.047	−.322***	−.264***	−.160**	−.145*	−.269***	—
8. Household income	−.099	.167**	.104	.137*	.139*	.159**	−.091

*$p < .05$. **$p < .01$. ***$p < .001$.

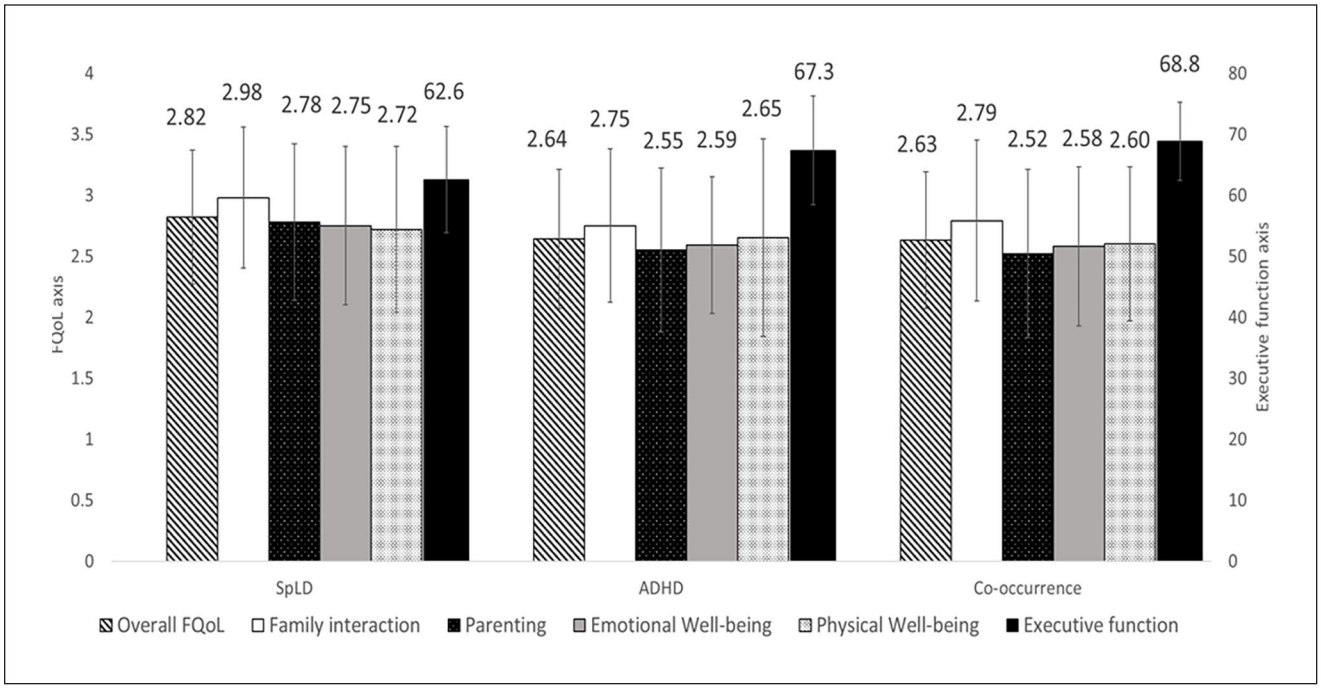


Figure 2. The Mean of FQoL and BRIEF T-Scores for the Three Groups of Children.
Note. Error bars are standard deviations. FQoL = family quality of life; BRIEF = Behavior Rating Inventory of Executive Function (LeJeune et al., 2010); abbreviated version.

severe impairments in EF were associated with lower FQoL scores. Importantly, when accounting for the mediating effect of EF, the direct effects of group classification on Family Interaction ($p = .94$ and $p = .27$) and Parenting ($p = .69$ and $p = .22$) and other moderation pathways of household income were not significant (see Figure 1B). In the Family Interaction model, the moderated mediation index for ADHD vs Co-occurrence groups was 0.09, 95% confidence interval (CI) = [0.01, 0.19], but it was not significant for SpLD versus Co-occurrence, which was 0.01, 95% CI = [−0.05, 0.06]. Similarly, in the Parenting model, the moderated mediation index for ADHD versus Co-occurrence was 0.08, 95% CI = [0.01, 0.16], while the confidence interval did not exclude 0 for SpLD versus Co-occurrence at 0.004, 95% CI = [−0.04, 0.05]. Taken together, household income moderated the ADHD versus Co-occurrence group effect on Family Interaction and Parenting through the effect of EF, while SpLD versus Co-occurrence group difference on Family Interaction and Parenting was mediated by EF but not dependent on household income.

Discussion

The current study investigated the FQoL of 217 families with children with ADHD, SpLD, or their co-occurrence in Hong Kong. Specifically, we compared the FQoL of these families with that of the general population and explored how these conditions, in conjunction with household income affects a child's EF and overall FQoL. Based on our

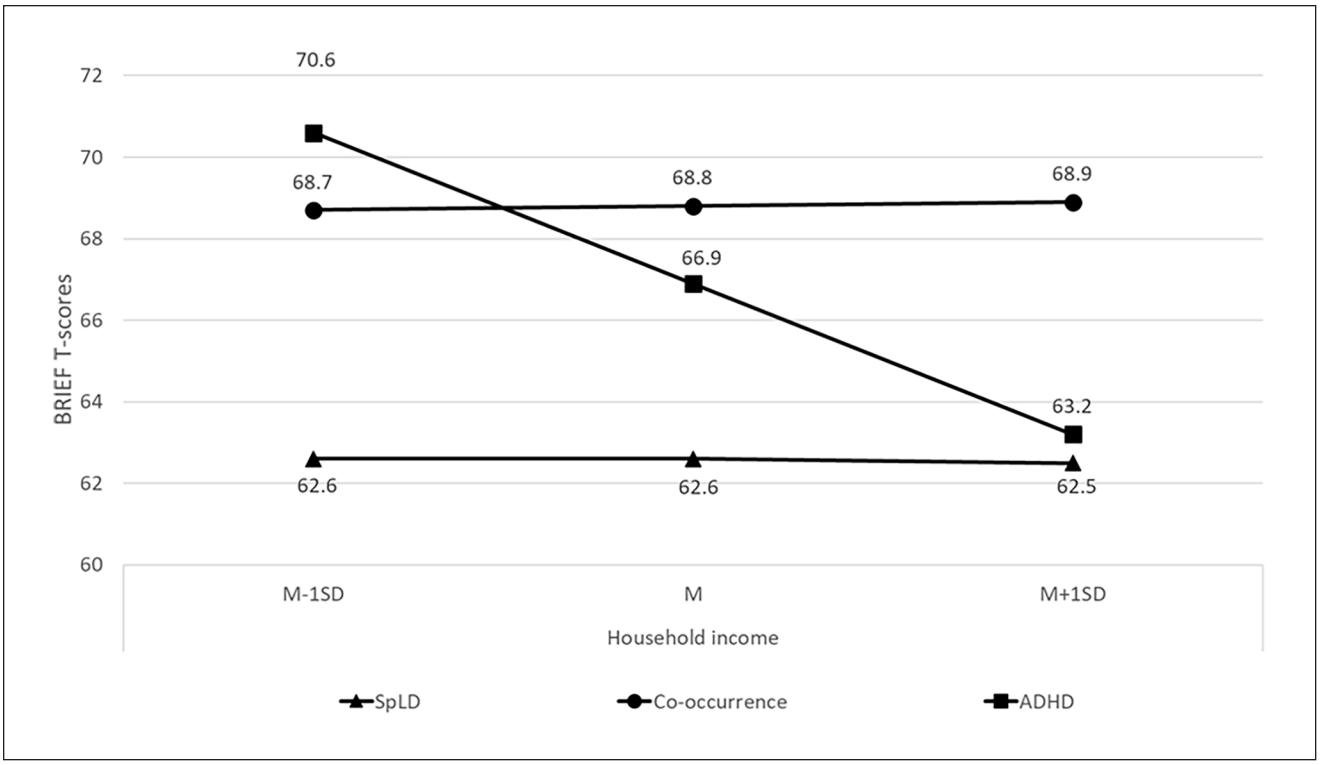


Figure 3. BRIEF T-Scores at Low, Medium, and High Family Income Levels Across Groups.
Note. SpLD = specific learning disabilities; ADHD = attention-deficit/hyperactivity disorder; BRIEF = Behavior Rating Inventory of Executive Function (LeJeune et al., 2010); abbreviated version.

results, we found that the families in our sample generally reported low FQoL, with negative to neutral scores on the Beach Center Family Quality of Life Scale (BC-FQoL), averaging below 3 on a 5-point scale. In contrast, reference surveys by the Hong Kong Family Welfare Society (HKFWS) in 2017 and 2018 found significantly higher BC-FQoL mean scores of 3.69 to 4.08 among the general population (n = 1,033 and 1,003, respectively; HKFWS, 2017, 2018). Differences in social environments, particularly family SES—a variable known to positively correlate with FQoL (Wang et al., 2004)—likely contributed to these discrepancies. While more than half of the HKFWS participants reported a monthly income exceeding HK$30,000, less than 25% of our sample fell into this income bracket. Therefore, our findings should be considered in light of the sample's demographic characteristics.

Beyond sample differences, the stark disparity in the reported FQoL between our sample and the community average may be due to the unique circumstances associated with caring for children with special educational needs. Existing studies have reported a negative correlation between FQoL and children with ADHD, SpLD, and other special educational needs (e.g., Romaniuk et al., 2022; Xiang et al., 2009; Xu et al., 2025). It is equally important to recognize the mutual influence of family members on overall family well-being. Families may respond to children's learning disabilities with great efforts and financial investments in behavioral, emotional, and social management, potentially resulting in diminished FQoL (Dizdarevic et al., 2022; Ginieri-Coccossis et al., 2013; Xiang et al., 2009). While corroborating results from Xu et al. (2025), this study found relatively low FQoL among families with children diagnosed with SpLD compared with community averages. This could be due to parental distress related to academic concerns or the social stigma surrounding SpLD and lower academic performance in the competitive educational atmosphere in Asia (R. C. H. Chan et al., 2020; Deater-Deckard et al., 2012). It is noteworthy that the lowest FQoL scores were reported for families with children in the ADHD and co-occurrence groups, suggesting that the additional challenges associated with ADHD, such as behavioral issues and attentional difficulties, may further exacerbate family stress and reduce quality of life. This finding expands on the existing research on FQoL in children with developmental disabilities by highlighting the varying impacts of different conditions.

The Mediation Effect of Executive Dysfunction

A central aim of the current study is to examine whether and how the child's EF may mediate the relationship between their learning disabilities status and FQoL. Previous research has reported that comorbid conditions may exacerbate executive dysfunction, which, in turn, impacts the

daily activities of children and their families, ultimately affecting FQoL (Dajani et al., 2016). Our study partially aligns with these findings, with group comparisons revealing that children with co-occurring SpLD and ADHD exhibited elevated executive dysfunction compared with those with SpLD alone. Notably, however, when family income was not taken into account, the executive dysfunction levels observed in children with co-occurring SpLD and ADHD did not significantly differ from those with ADHD alone. One possible explanation was that the number of co-occurring special educational needs was higher on average in the ADHD group compared with the SpLD group (0.7 and 0.3, respectively). As many different special educational needs, such as autism spectrum disorder, may influence EF, polymorbidity in diagnostic background may be correlated with greater reported executive dysfunction. However, this pattern was also consistent with prior studies suggesting that comorbidity does not necessarily lead to more severe executive dysfunction compared with single diagnoses of SpLD or ADHD (Fernández-Andrés et al., 2019; Moura et al., 2016). Instead, learning disabilities such as SpLD or ADHD may be associated with distinct profiles of EF, which may relate to FQoL in different ways. For instance, more pronounced executive dysfunction could exacerbate some children's behavioral issues, for example, noncompliance or social problems, leading to stricter parental discipline and impacting family interactions (Deater-Deckard et al., 2012). Executive dysfunction in ADHD may also present as the child's difficulties initiating or persisting in tasks, requiring more parenting efforts (R. C. H. Chan et al., 2020). A crucial finding of this study was that the child's executive dysfunction slightly impaired specific domains of FQoL in the family interaction and parenting domains.

The Moderation Effect of Household Income

Our study also identified a moderating effect of family income on the relationship between group status (ADHD vs. co-occurrence) and executive dysfunction. Although the overall EF levels were similar between ADHD and co-occurrence groups, for children with ADHD alone, higher family income appeared to offer a protective buffer against the adverse effects of executive dysfunction on family well-being. This result highlighted the role of household income in mitigating the impact of ADHD, as suggested by previous research (Guajardo et al., 2009). Contrary to our expectations, however, the impact of co-occurring learning disabilities on EF did not vary significantly across different household income levels. Meanwhile, the SpLD group demonstrated lower levels of executive dysfunction compared with the co-occurrence group, also regardless of household income levels. This overall pattern aligned with previous research suggesting that comorbid SpLD and ADHD can result in distinct patterns of executive dysfunction compared with SpLD alone (Bental & Tirosh, 2007; Poon & Ho, 2014; Willcutt et al., 2005). The co-occurrence group may exhibit a broader range of executive dysfunctions which might reduce the protective benefits typically associated with higher household income. However, as our sample primarily consisted of families from lower SES backgrounds, this may have influenced the observed effects (Blair, 2010; Duncan et al., 2017). Future research should explore this relationship further, taking into account the complexity of EF in children with both SpLD and ADHD.

The significance of SES is underscored by the broader context of income inequality in Hong Kong. In Hong Kong, approximately 20% of the population is estimated to be living in poverty (Choi et al., 2022). In situations where children from low-SES families have more complex needs, such as co-occurring special educational needs, these families often lack sufficient social or financial support, leading to increased parenting stress (Duncan et al., 2017; Guajardo et al., 2009). Moreover, it is important to consider the timing of our study, which was conducted during the pandemic—a period when the effects of SES were likely intensified. The pandemic caused the unemployment rate in Hong Kong to peak at 6.5% by the end of 2020 and further exacerbated poverty levels (Choi et al., 2022). As a result, many children struggled to adapt to sudden disruptions in their daily lives and confinement at home, which could have led to increased anxiety or depression (Fogel et al., 2022). This context may have placed an even greater strain on families caring for children with SpLD or ADHD, particularly in terms of family health and resources (Dhiman et al., 2020).

Implications

This study represents the first comprehensive exploration of FQoL among children with SpLD, ADHD, and their co-occurrence in Hong Kong, paving the way for future research. Our findings revealed a reliable connection between a child's EF and their FQoL, which has practical implications for service providers and policymakers aiming to cater to the unique needs of these families. Given our findings that EF plays a role in the FQoL for this population, clinicians should consider understanding children's EF at an early stage. This might involve early identification and support, alongside caregiver guidance, to potentially enhance family relationship and child adaptation in home, school, and community settings. Policymakers could consider implementing comprehensive support measures for families with children with co-occurring SpLD and ADHD, as these families face ongoing challenges related to managing behavioral, emotional, and attentional difficulties, which persist across SES levels. By providing these families with a range of resources, such as educational programs, counseling services, and community support, the FQoL of

these families may show a corresponding increase. Furthermore, our research underscores the importance of adopting a family-centered model in education and rehabilitation services. This model could include parental education that involves strategies to improve the quality of family interactions and parenting skills, and timely services and relevant support to alleviate parental and caregiver stress, ultimately enhancing the FQoL of these families.

Limitations and Future Directions

Several limitations in this study warrant consideration. First, it is essential to consider the timing of this study, which coincided with the COVID-19 pandemic and led to restrictions on the data collection process. For instance, the study mainly relied on caregiver self-reports on the child's overall EF performance, which may introduce the possibility of respondent bias. Future studies should include task-based EF performance data for more comprehensive assessments that can separate components of EF, such as working memory, inhibitory control, and cognitive flexibility. Second, this study employed a cross-sectional survey, limiting the ability to draw causal inferences. Third, despite our best efforts, the group sizes and co-occurring special educational needs numbers were not matched for the three groups of participants. Finally, although surveys conducted by HKFWS reported the FQoL scores of the general population in 2017 and 2018, the lack of control group data from families with similar demographic backgrounds makes direct comparisons challenging. Future research could employ longitudinal or experimental designs to investigate these variables' causal relationships thoroughly.

Conclusion

This study significantly enhances our understanding of FQoL among children with SpLD, ADHD, and their co-occurrence in Hong Kong. It highlights how executive dysfunction may affect family interaction and parenting within these groups. In addition, the study shows that family income plays a crucial role in moderating the child's executive function in the ADHD group, relative to the co-occurrence group. These findings emphasize the need for future interventions and family-based support systems to improve the overall well-being of these families.

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Credit Author Statement

D.L. performed data curation, formal analysis, visualization, and writing of the original draft. K.P. performed funding acquisition, investigation, project administration, and reviewed the writing. Y.Y.N. performed conceptualization, formal analysis, methodology, and writing and editing.

Availability of Data and Materials

The data and materials are not publicly archived because they pertain to a vulnerable population. The dataset analyzed may be obtained from the corresponding author on reasonable request.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Ethics Approval

The study was performed in accordance with the Declaration of Helsinki with approval of the Education University of Hong Kong Human Research Ethics Committee (#A2018-2019-0261).

Consent to Publication

All data were anonymous and not linked to individuals.

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